



Dirt cheap malware on Macs

A malware by the name Xloader can be licensed by criminals for as low as \$49. <https://dgit.in/aug21-05>



WiiBoy Color

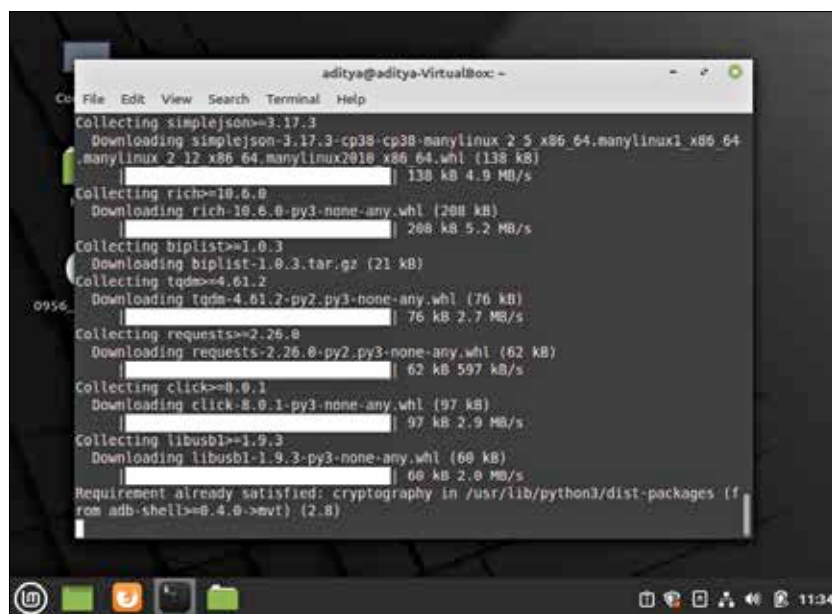
A modded Wii in a gameboy's outfit! Console modder GingerOfMods delivers an amazing experience. <https://dgit.in/aug21-06>

As Apple pushes out the security updates to users directly, while the Android patches for vulnerabilities are staggered through device manufacturers as well as telecom service providers, there are more droids with vulnerabilities that both the malware use for infection and the forensic tools use for the purposes of data extraction. This also

of the device itself, to get some kinds of data in realtime from the app such as messages, calls and the data used by apps. This can be a dump of all the messages and call logs. A file system dump of a device is a copy of all the data and information in a device, as it is stored. Then there is a physical extraction, which is an exacting copy of the memory of the device, allowing

file size, but without all the data needed for MVT to scan. You will need to enter in a password that you have to remember for that particular encrypted backup, and use the passcode to unlock the phone tethered to the computer during the backing up process. Once you have the backup ready, you have to access it through a CLI where you can actually use MVT. There are several ways this can be done. The best way to do this is with a Mac or Linux machine, although a Windows machine with Windows Subsystem for Linux or VirtualBox works just as fine. Another quick fix is to use a USB drive with sufficient space on it and install Linux on it, flashing a Linux Mint ISO using the Balena Etcher is a good option. One of the simple ways to load the backup into a Linux instance is to use an ISO file and mount it as a disc. Either way, once you have access to the backup in a Linux terminal, you have to install python, then the MVT, and scan the backups. For Android devices, the methodology is simple, connect the device to the machine with a terminal, put it in developer mode, and let the software do the work. MVT flags detections of Pegasus at this point, but at that point it is a bit too late. When it comes to spyware, the principle of "prevention is better than cure" applies.

So what does a user do when an infection is detected? Krishnapur tells us, "Once a device is taken over by Pegasus, there's very little a victim can do because it is a zero-day attack with user-less interaction, in-memory (stealth), self-destruction, encryption and persistence capabilities – all phenomenal features that make it extremely hard to detect. Given that Pegasus exploits previously unknown loopholes in apps, devices cannot be protected by traditional means. A patch is needed to fix the vulnerability and till it is known and patched, it is that window of opportunity that is being exploited. Another aspect is that since it has significant persistence capabilities where it lives



Installing the Mobile Verification Toolkit

makes it harder to detect Pegasus on an Android device, so the toolkit uses the approach of scanning the messages for suspicious links, and the installed apps themselves. The toolkit posted on the Amnesty International website is developed in collaboration with the researchers from Citizen Lab who have been tracking this spyware for years.

To go ahead and check your own device for signs of tools such as Pegasus, it is necessary to do a forensic analysis of the phone, and extract the data. For the purposes of detecting cutting edge spyware that uses new vulnerabilities, it is necessary to audit all the data on the device, the bits sent between apps, the bits sent to the cloud, and all the bits on the flash memory, even the traces of deleted files. A logical extraction uses the internal language

for recovery of deleted files such as notes, recordings and images. If the device uses password protection and encryption, then a lot of the data would not be accessible without the passcodes to get past the locked data. The simplest way to test an iOS device is to use an encrypted backup of the device, which is run through the command line tool available on GitHub (<https://github.com/mvt-project/mvt>). To make the backup, you have to go through the painful procedure of installing iTunes on the machine, and making an encrypted backup of the device from File > Devices > Backup. For the purposes of detection using the Mobile Verification Toolkit (MVT), it is better to encrypt the backup rather than not, which actually stores the encrypted data from the phone on the machine. Without this option checked, there is a smaller